

| Question |       |       |       | Marking details                                                                                                                                                                                                                                                                                          | Marks available |       |      |       |          |      |   |   |          |       |       |       |  |  |   |   |  |   |
|----------|-------|-------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------|------|-------|----------|------|---|---|----------|-------|-------|-------|--|--|---|---|--|---|
|          |       |       |       |                                                                                                                                                                                                                                                                                                          | AO1             | AO2   | AO3  | Total | Maths    | Prac |   |   |          |       |       |       |  |  |   |   |  |   |
| 9.       | (a)   |       |       | Alpha – <u>helium nucleus</u> (1) Treat 2 protons and 2 neutrons as neutral<br>Beta – [fast moving] electron (1)<br>Gamma – em wave (1)                                                                                                                                                                  | 3               |       |      | 3     |          |      |   |   |          |       |       |       |  |  |   |   |  |   |
|          | (b)   | (i)   |       | $\frac{18}{30} = 0.6$ [counts per second]                                                                                                                                                                                                                                                                |                 | 1     |      | 1     | 1        | 1    |   |   |          |       |       |       |  |  |   |   |  |   |
|          |       | (ii)  |       | Measure for longer (1)<br>Repeat (1)                                                                                                                                                                                                                                                                     |                 |       | 2    | 2     |          | 2    |   |   |          |       |       |       |  |  |   |   |  |   |
|          |       | (iii) |       | [More] radon (1)<br>Due to granite or different types of rock (1)<br><br><b>Alternative:</b><br>[More] cosmic rays (1)<br>Due to high altitude (1)                                                                                                                                                       | 2               |       |      | 2     |          |      |   |   |          |       |       |       |  |  |   |   |  |   |
|          | (c)   | (i)   |       | <table border="1"><tr><td></td><td>Alpha</td><td>Beta</td><td>Gamma</td></tr><tr><td>Source 1</td><td>N</td><td>N</td><td>Y</td></tr><tr><td>Source 2</td><td>Y (1)</td><td>Y (1)</td><td>N (1)</td></tr></table><br>Award 1 mark for source 1 row being correct<br>Accept ticks and crosses for Y and N |                 | Alpha | Beta | Gamma | Source 1 | N    | N | Y | Source 2 | Y (1) | Y (1) | N (1) |  |  | 4 | 4 |  | 4 |
|          | Alpha | Beta  | Gamma |                                                                                                                                                                                                                                                                                                          |                 |       |      |       |          |      |   |   |          |       |       |       |  |  |   |   |  |   |
| Source 1 | N     | N     | Y     |                                                                                                                                                                                                                                                                                                          |                 |       |      |       |          |      |   |   |          |       |       |       |  |  |   |   |  |   |
| Source 2 | Y (1) | Y (1) | N (1) |                                                                                                                                                                                                                                                                                                          |                 |       |      |       |          |      |   |   |          |       |       |       |  |  |   |   |  |   |